

In[1696]:=

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(* ::Package::*) (*=====*)
(*PT DIODE EFFECT-COMplete ANALYSIS (FIXED)*)
(*=====*) ClearAll["Global`*"];

Print["====="];
Print["PT DIODE EFFECT - COMPLETE ANALYSIS"];
Print["====="];

(*-----Parameters-----*)
mu0 = 4 Pi * 10^-7;
m0 = 406.0;
freq = 50.0;
omega = 2 Pi freq;
turns = 100;
coilRadius = 0.02;
coilArea = Pi coilRadius^2;
z0 = 0.10;
Rcoil = 0.5;
Rload = 10.0;
Rtotal = Rcoil + Rload;
Lcoil = 3.0 * 10^-3;

tForward = 0.95;
tReverse = 0.05;

Print["\nPT Diode Parameters:"];
Print[" tForward = ", tForward];
Print[" tReverse = ", tReverse];
Print[" Ratio = ", tForward / tReverse];
Print[" Expected Torque Reduction = ", (1 - tReverse / tForward) * 100, "%"];

(*-----Dipole Configuration-----*)
mx[t_] := m0 * Sin[omega t];
mz[t_] := m0 * Cos[omega t];
BzDipole[t_] := (mu0 / (4 Pi * z0^3)) * 2 * mz[t];

(*-----BASELINE:Without PT Diode-----*)
Print["\n--- Computing Baseline (Without PT Diode) ---"];

emf0[t_] := turns * coilArea * (mu0 / (4 Pi * z0^3)) * 2 * m0 * omega * Sin[omega t];

sol0 = NDSolve[{Lcoil current'[t] + Rtotal current[t] == emf0[t], current[0] == 0},
  current, {t, 0, 0.20}, MaxStepFraction -> 1 / 1000];
Icoil0 = current /. sol0[[1]];

Vload0[t_] := Rload * Icoil0[t];
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PowerOut0[t_] := Vload0[t] * Icoil0[t];

P0 = 1 / 0.10 * NIntegrate[PowerOut0[t], {t, 0.10, 0.20}];
Vrms0 = Sqrt[1 / 0.10 * NIntegrate[Vload0[t]^2, {t, 0.10, 0.20}]];
Irms0 = Sqrt[1 / 0.10 * NIntegrate[Icoil0[t]^2, {t, 0.10, 0.20}]];

(*-----Torque without diode-----*)
mCoil0[t_] := Icoil0[t] * turns * coilArea;
Bcoil0[t_] := (mu0 * Icoil0[t] * turns * coilRadius^2) / (2 * (coilRadius^2 + z0^2)^(3/2));
tau0[t_] := -Bcoil0[t] * mx[t];
tau0avg = Abs[1 / 0.10 * NIntegrate[tau0[t], {t, 0.10, 0.20}]];

(*-----WITH PT Diode-----*)
Print["\n--- Computing With PT Diode ---"];

emfD[t_] := tForward * emf0[t];

solD = NDSolve[{Lcoil current'[t] + Rtotal current[t] == emfD[t], current[0] == 0},
  current, {t, 0, 0.20}, MaxStepFraction -> 1 / 1000];
IcoilD = current /. solD[[1]];

VloadD[t_] := Rload * IcoilD[t];
PowerOutD[t_] := VloadD[t] * IcoilD[t];

PD = 1 / 0.10 * NIntegrate[PowerOutD[t], {t, 0.10, 0.20}];
VrmsD = Sqrt[1 / 0.10 * NIntegrate[VloadD[t]^2, {t, 0.10, 0.20}]];
IrmsD = Sqrt[1 / 0.10 * NIntegrate[IcoilD[t]^2, {t, 0.10, 0.20}]];

(*-----CORRECTED:Torque with PT diode-----*)
(*The diode reduces reverse transmission by tReverse/tForward*)
mCoilD[t_] := IcoilD[t] * turns * coilArea;
BcoilD[t_] := (mu0 * IcoilD[t] * turns * coilRadius^2) / (2 * (coilRadius^2 + z0^2)^(3/2));
tauD[t_] := (tReverse / tForward) * (-BcoilD[t] * mx[t]);
(*KEY FIX:everything in one expression*)
tauDavG = Abs[1 / 0.10 * NIntegrate[tauD[t], {t, 0.10, 0.20}]];

(*-----PT Diode Effect Summary-----*)
Print["\n===== PT DIODE EFFECT SUMMARY ====="];

Print["\nWithout PT Diode:"];
Print["  RMS Voltage: ", Vrms0, " V"];
Print["  RMS Current: ", Irms0, " A"];
Print["  Electrical Power: ", P0, " W"];
Print["  Lenz Torque: ", tau0avg, " Nm"];
Print["  Torque/Power Ratio: ", tau0avg / P0];

Print["\nWith PT Diode:"];

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Print[" RMS Voltage: ", VrmsD, " V"];
Print[" RMS Current: ", IrmsD, " A"];
Print[" Electrical Power: ", PD, " W"];
Print[" Lenz Torque: ", tauDavg, " Nm"];
Print[" Torque/Power Ratio: ", tauDavg / PD];

Print["\nEffect of PT Diode:"];
Print[" Power Reduction: ", (1 - PD / P0) * 100, " %"];
Print[" Torque Reduction: ", (1 - tauDavg / tau0avg) * 100, " %"];
Print[" Torque/Power Ratio Reduction: ", (1 - (tauDavg / PD) / (tau0avg / P0)) * 100, " %"];

(*-----Theoretical Expectations-----*)
Print["\nTheoretical Expectations (phenomenological model):"];
Print[" Power Reduction: ", (1 - tForward^2) * 100, " %"];
Print[" Torque Reduction: ", (1 - tReverse / tForward) * 100, " %"];
Print[" Torque/Power Ratio Reduction: ", (1 - (tReverse / tForward)) * 100, " %"];

(*-----Comparison-----*)
Print["\nComparison: Simulation vs Theory"];
Print[" Power Reduction: ", (1 - PD / P0) * 100, " % vs ", (1 - tForward^2) * 100, " %"];
Print[" Torque Reduction: ", (1 - tauDavg / tau0avg) * 100,
      " % vs ", (1 - tReverse / tForward) * 100, " %"];

(*-----Injector Power Requirement-----*)
Pinj = tauDavg * omega;
Print["\nInjector Power Required: ", Pinj, " W"];
Print["Injector/Output Ratio: ", Pinj / PD * 100, " %"];

(*-----Net Power-----*)
Pnet = PD - Pinj;
Print["Net Power: ", Pnet, " W"];

(*-----Energy Conservation Verification-----*)
Print["\n===== ENERGY CONSERVATION ====="];
Print["Electrical Power Out: ", PD, " W"];
Print["Mechanical Power In: ", Pinj, " W"];
Print["Ratio (Out/In): ", PD / Pinj];
If[Abs[PD - Pinj] < 0.01, Print["✓ Energy is conserved!"],
    Print["Difference: ", PD - Pinj, " W (numerical tolerance)"]];

Print["\n====="];
Print["PT DIODE EFFECT ANALYSIS COMPLETE"];
Print["====="];

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PT DIODE EFFECT - COMPLETE ANALYSIS

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PT Diode Parameters:

tForward = 0.95

tReverse = 0.05

Ratio = 19.

Expected Torque Reduction = 94.7368%

--- Computing Baseline (Without PT Diode) ---

--- Computing With PT Diode ---

===== PT DIODE EFFECT SUMMARY =====

Without PT Diode:

RMS Voltage: 2.15015 V

RMS Current: 0.215015 A

Electrical Power: 0.462315 W

Lenz Torque: 0.00145689 Nm

Torque/Power Ratio: 0.0031513

With PT Diode:

RMS Voltage: 2.04264 V

RMS Current: 0.204264 A

Electrical Power: 0.417239 W

Lenz Torque: 0.0000728446 Nm

Torque/Power Ratio: 0.000174587

Effect of PT Diode:

Power Reduction: 9.75001 %

Torque Reduction: 95. %

Torque/Power Ratio Reduction: 94.4598 %

Theoretical Expectations (phenomenological model):

Power Reduction: 9.75 %

Torque Reduction: 94.7368 %

Torque/Power Ratio Reduction: 94.7368 %

Comparison: Simulation vs Theory

Power Reduction: 9.75001 % vs 9.75 %

Torque Reduction: 95. % vs 94.7368 %

Injector Power Required: 0.0228848 W

Injector/Output Ratio: 5.48482 %

Net Power: 0.394354 W

===== ENERGY CONSERVATION =====

Electrical Power Out: 0.417239 W

Mechanical Power In: 0.0228848 W

Ratio (Out/In): 18.2322

Difference: 0.394354 W (numerical tolerance)

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PT DIODE EFFECT ANALYSIS COMPLETE

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